

7.2 QUANTUM PHYSICS P2, P3&P5 QUESTIONS

Compiled by: R. Muza

Q1:NOV 2002 P2

- 7 (a) Give an expression for E , the energy of a photon in terms of f , its frequency, and h the Planck constant.

[1]

- (b) Some of the energy levels of an atom of one substance are shown in Fig. 7.1 below.

energy in joules $\times 10^{-19}$	energy level n
0.0	∞
-2.56	4
-5.92	3
-8.80	2
-16.64	1

Fig. 7.1

- (i) How much energy is required to raise an electron from $n = 1$ to $n = 4$.

energy = _____

[1]

- (ii) Calculate the wavelength of the radiation that is given off by the electron.

wavelength = _____

[2]

7.2 QUANTUM PHYSICS P2, P3&P5 QUESTIONS

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- (iii) What part of the electromagnetic spectrum does this radiation belong to?

_____ [1]

Q2:JUN 2014 P2

- 7 (a) Define *the photoelectric effect*.

_____ [2]

- (b) Describe how the following facts from the photoelectric experiment contradict the wave theory of light:

- (i) *the existence of a photoelectric threshold frequency*

- (ii) *the direct proportionality between the intensity and the photoelectric current*

- (iii) *instantaneous emission of electrons*

[3]

- (c) Light of wavelength 400 nm is incident on a metal surface with a work function of 3.6×10^{-19} J.

Calculate

- (i) the energy of the incident light,

- (ii) the maximum kinetic energy of the emitted electron.

kinetic energy = _____

[4]

Q3:NOV 2013 P2

- 8 (a) Define the following terms as they are used in the photoelectric effect:

- (i) *photon*

- (ii) *threshold wavelength*

- (iii) *stopping potential*

[3]

- (b) Light of wavelength 5.89×10^{-7} m is incident on a metal surface.

Calculate for a photon of this wavelength, the

- (i) energy, E ,

(ii) momentum, p .

$$p = \underline{\hspace{2cm}}$$

[4]

Q4:NOV 2012 P2

9 Fig.9.1 shows a photoemissive cell circuit.

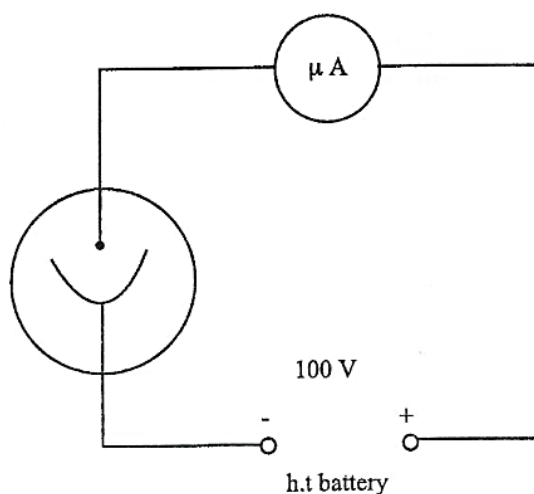


Fig. 9.1

- (a) Draw a circuit design, incorporating part or the whole of Fig.9.1, for a 12 V burglar alarm which rings if a light beam is broken by an intruder.

[2]

- (b) A UV and **not** an infrared source of light might be a good idea for the burglar alarm using a photocell with a cathode-coating work function of 3 eV.

Explain. _____

[4]

Q5:JUN 2012 P2

- 6 The work function of a metal is 2.0 eV. Monochromatic light of wavelength $5.0 \times 10^{-7} \text{ m}$ is used to illuminate the metal.

(a) Explain the term *work function*.

[1]

(b) Calculate

(i) the threshold wavelength, λ ,

$$\lambda = \underline{\hspace{2cm}}$$

(ii) the maximum kinetic energy of photoelectrons,

$$\text{maximum kinetic energy} = \underline{\hspace{2cm}}$$

7.2 QUANTUM PHYSICS P2, P3&P5 QUESTIONS

Compiled by: R. Muza

(iii) the stopping potential.

stopping potential = _____

[6]

(c) Suggest the effects of using an intense light of wavelength 75 nm.

[2]

Q6:JUN 2010 P2

8 (a) Define

1. *work function,*

2. *threshold frequency.*

[2]

- (b) In a photoelectric emission experiment, a metal surface in an evacuated tube is illuminated with monochromatic light.

The experiment is repeated with light of the same wavelength but of double the intensity. State how this affects

1. the energy of a photon,

2. maximum kinetic energy of photoelectrons,

3. work function of the metal,

4. photoelectric current.

 [4]

Q7:JUN 2008 P2

- 10 In a photoelectric experiment the circuit in Fig. 10.1 was used to produce the graph shown in Fig. 10.2:

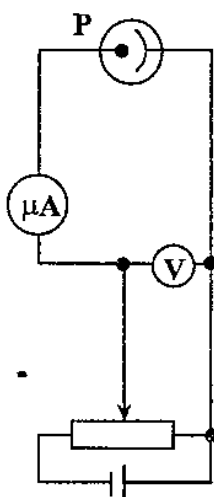


Fig. 10.1

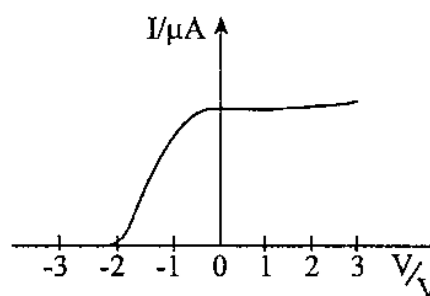


Fig.10.2

- (a) Explain why the current stays constant if the cell in Fig. 10.1 is reversed and P is made more and more positive.

 [1]

- (b) The photoelectric current falls when the potential difference V is made more and more negative.

Explain.

[1]

- (c) The wavelength of light used was 500 nm.

Calculate the work function of the metal.

work function = _____ [2]

- (d) On Fig. 10.2 sketch the curves you would expect for

- (i) increased light intensity. Label it R,
 (ii) increased light frequency. Label it S. [2]

Q8:NOV 2016 P2

- 9 (a) Given the photoelectric equation:

$$hf = \phi + E_k$$

State the physical quantity represented by

- (i) hf , _____
 (ii) ϕ , _____
 (iii) E_k , _____

[3]

- (b)** A metal surface of threshold wavelength 550 nm is irradiated with blue light of wavelength 450 nm.

Calculate the

- (i)** maximum kinetic energy of the emitted photoelectrons,
- (ii)** reverse voltage required to stop the emitted photoelectrons from reaching the anode.

[4]

- (c)** State, with a reason, the effect on the quantity calculated in **b(ii)** of doubling the

- (i)** intensity of incident light, _____

- (ii)** wavelength. _____

[4]

Q9: NOV 2014 P2

- 10 (a) The terms *work function* and *threshold wavelength* are used in the description of the photoelectric effect.

Define

- (i) *work function*,

- (ii) *threshold wavelength*.

[2]

- (b) In an α -scattering experiment using gold foil, the α -particles have an energy of 5.6 MeV.

Given that the mass of the α -particle is 6.7×10^{-27} kg and its charge is $+3.2 \times 10^{-19}$ C, calculate

- (i) the velocity of the α -particles,

velocity = _____

- (ii) the distance of closest approach between the centres of a gold nucleus and an α -particle. (proton number of gold = 79).

distance = _____

[4]

Q10:JUN 2017 P2

7 (a) Explain the following terms:

(i) *photon*

(ii) *wave particle duality*

[2]

(b) The photoelectric effect provides evidence for the particulate nature of electromagnetic radiation.

State **three** experimental observations that support this conclusion.

1.

2.

3.

[3]

- (c) Calculate the wavelength of a beam of electrons which was accelerated through a potential difference of 3 600 V.

wavelength = _____

[3]

- (d) Fig. 7.1 shows sunlight rays directed to an enclosed hydrogen gas.

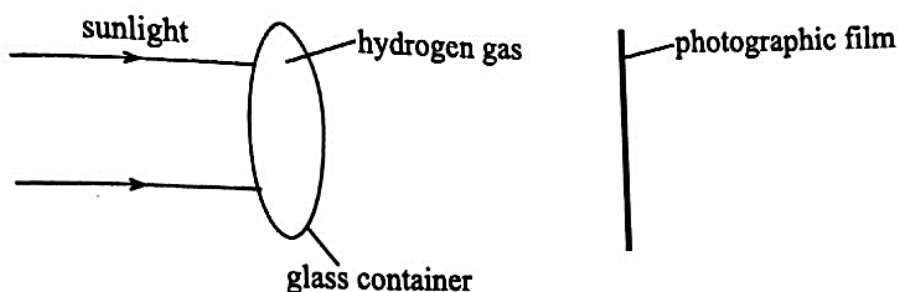


Fig. 7.1

- (i) State the type of spectrum observed on the photographic film.

- (ii) Explain, with a reason, how the spectrum in (i) is produced.

[3]

Q11:JUN 2015 P2

- 9 (a) Define, in words, the term

- (i) *photon*,

7.2 QUANTUM PHYSICS P2, P3&P5 QUESTIONS

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(ii) *work function,*

(iii) *threshold frequency,*

(iv) *photoelectric emission.*

[4]

(b) Calculate the energy of a photon of light of wavelength $5.89 \times 10^{-7} \text{ m}$.

energy = _____ [2]

Q12:NOV 2019 P2

- 6 (a) Wave particle duality suggests that light exhibits both wave and particle properties. If light falls upon a metal plate, photoelectrons are ejected.

Explain the observation from the viewpoint of

(i) wave theorists,

(ii) particulate theorists.

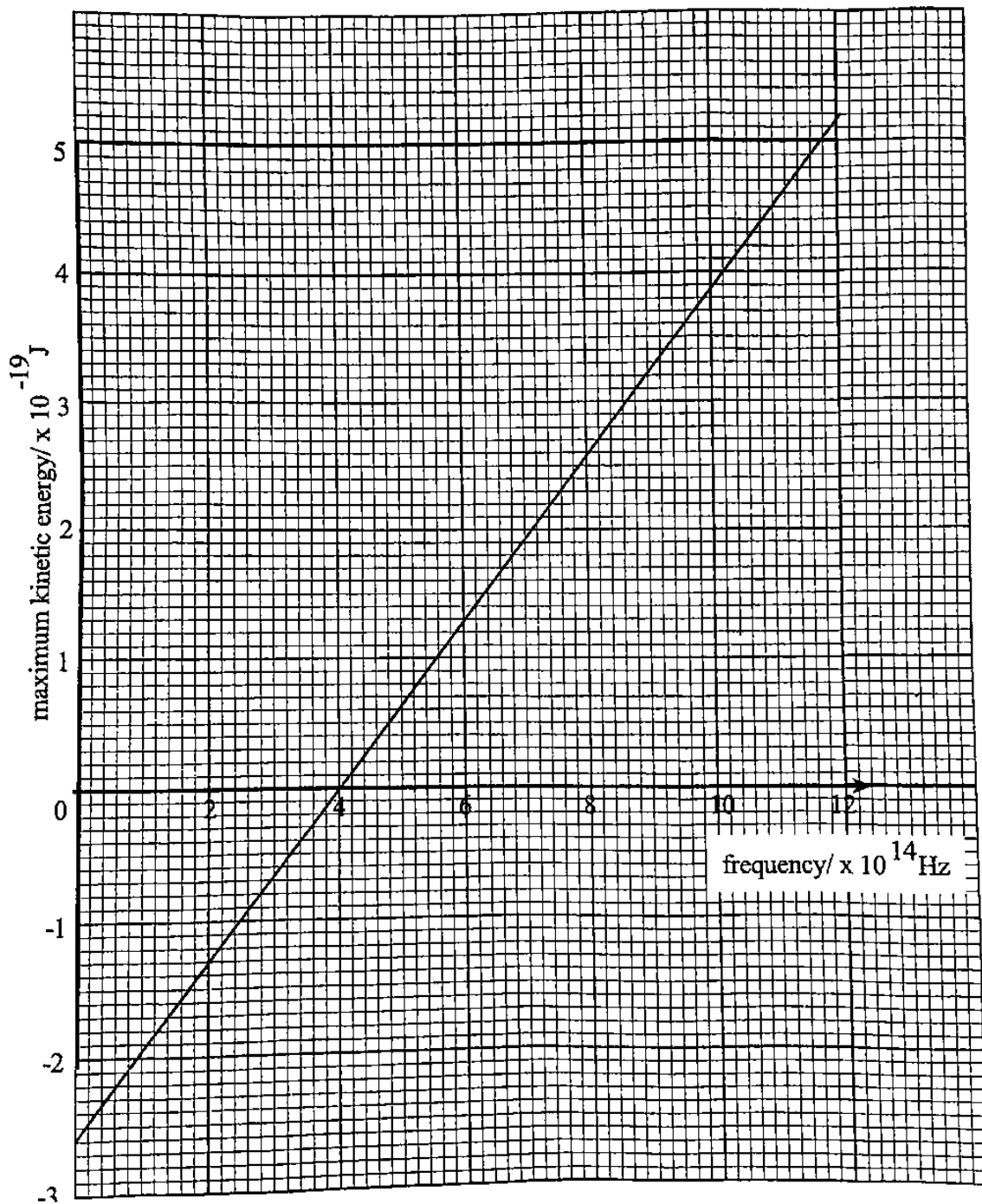
[4]

(b) State the experiments that provide evidence for the

(i) wave nature of particles,

(ii) particulate nature of waves.

(c) Fig. 6.1 shows a graph of maximum kinetic energy against frequency from a photoelectric experiment.



Deduce, from Fig. 6.1, the

(i) Plank's constant,

Plank's constant = _____

(ii) minimum frequency of radiation that causes photoelectric emission,

minimum frequency = _____

(iii) work function.

work function = _____ [4]

Q13:JUN 2010 P5

- 1 (c) (i) State the photoelectric emission equation.
- (ii) Fig. 1.2 shows a negatively charged gold leaf electroscope being illuminated by a red light of wave length 500 nm. The threshold frequency of potassium is 3.0×10^{14} Hz.

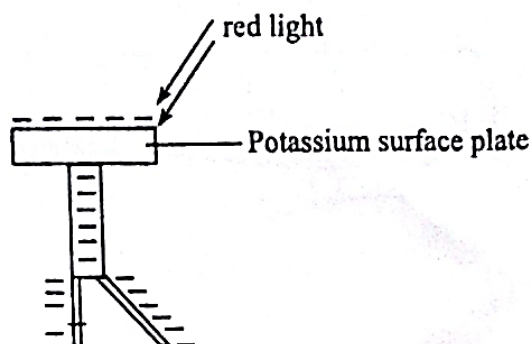


Fig. 1.2

- Describe and explain the effect of the red light on the electroscope.
- State the effect of increasing the brightness of the red light on the electroscope.
- Calculate the maximum velocity of the emitted electrons.

[8]

Q14:JUN 2014 P5

- 1 (d) (i) Explain the term *quantisation of charge*.
- (ii) Fig 1.4 shows the apparatus used in the Millikan's oil drop experiment.

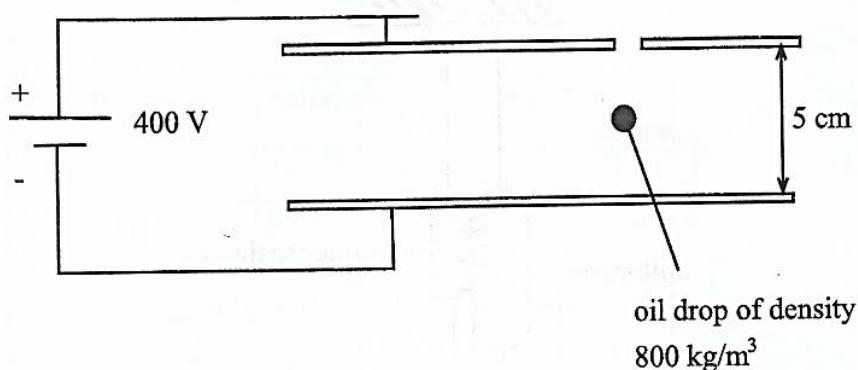


Fig. 1.4

A potential difference of 400 V was applied to just keep the oil drop at rest. The radius of the oil drop was measured and found to be 4×10^{-7} m.

7.2 QUANTUM PHYSICS P2, P3&P5 QUESTIONS

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1. Calculate the charge on the oil drop.
2. State and explain whether the value of the radius of the oil drop was accurately obtained.

[7]

Q15:JUN 2014 P5

- 5 (a) State the meaning of the term *workfunction*. [1]

- (b) A metallic surface is illuminated with monochromatic radiation which causes photoemissions.

State the factors that can increase the

- (i) number of photoelectrons emitted per second,
- (ii) maximum kinetic energy of photoelectrons.

[2]

- (c) Ultra-Violet radiation of wavelength 319 nm is illuminated on a metal surface and photoelectrons with a stopping potential of 0.22 V are produced.

Determine

- (i) the work function of the metal in eV,
- (ii) the maximum speed of photoelectrons,
- (iii) the threshold frequency of the metal.

[5]

- (d) Suggest a reason why most photoelectrons do not have the maximum kinetic energy. [1]

- (e) (i) State the de Broglie wavelength equation.

- (ii) Calculate the velocity of an α -particle for which the de Broglie wavelength is 6.4×10^{-15} m.

[3]

Q16:NOV 2011 P5

- 1 (c) (i) State what is meant by a *photon*.

- (ii) Explain how the photoelectric effect demonstrates the quantum nature of electromagnetic radiation.

[6]

Q17: NOV 2009 P5

- 1 (a) (i) State de Broglie's wave equation.
- (ii) A man of mass 50 kg runs through a door of dimensions 90×150 cm at a velocity of 7.9 ms^{-1} .
1. Calculate the wavelength associated with the man.
 2. State and explain whether there is a diffraction pattern observed.
 3. Explain briefly how the wave nature of particles can be demonstrated.
- [8]

Q18: JUN 2009 P5

- 3 (b) Fig. 3.2 shows some energy levels of a hydrogen atom (not to scale).

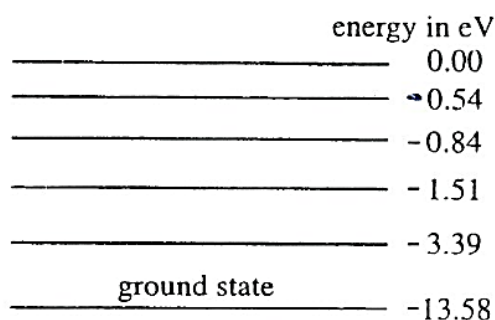


Fig. 3.2

- (i) Explain why the energy levels are
1. labelled with negative energies.
 2. the energy levels are not uniformly spaced.
- (ii) State the transition that will result in the emission of radiation of wavelength 436 nm. Justify your answer with a suitable calculation.
- (iii) Determine the frequency of electromagnetic radiation that can cause a transition from ground state to 0.00 eV
- [6]

Q19: NOV 2008 P5

- 1 (b) Fig. 1.2 shows the energy level diagram for a hydrogen atom.

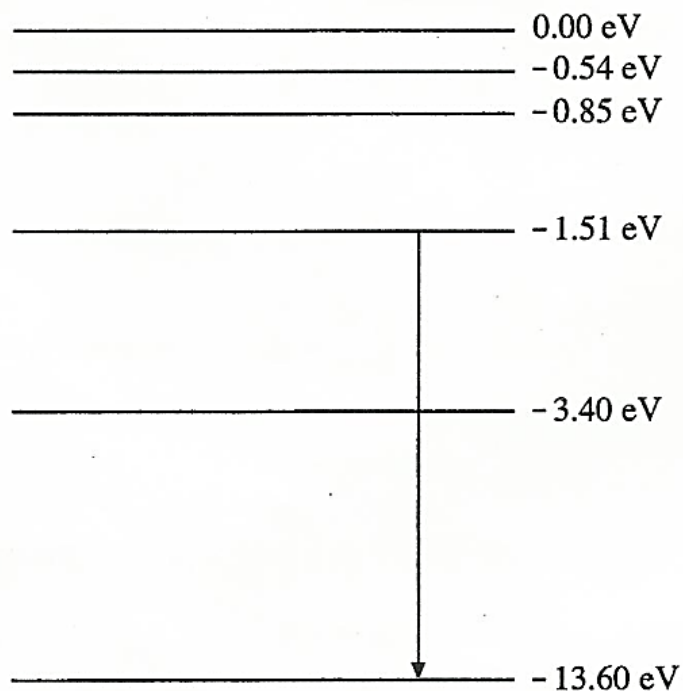


Fig. 1.2

- (i) Calculate the wavelength of the radiation emitted during the transition shown by the arrow in Fig. 1.2.
- (ii) State the region of the electromagnetic spectrum in which the emission occurs.
- (iii) Deduce the effects, if any, of illuminating the hydrogen atom by an X-ray beam of frequency 2.3×10^{16} Hz. [8]

Q20: JUN 2016 P5

- 1 (b) Fig. 1.2 shows electron energy levels of an atom.

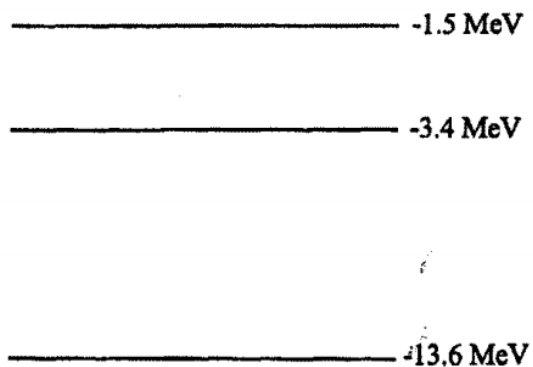


Fig. 1.2

7.2 QUANTUM PHYSICS P2, P3&P5 QUESTIONS

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- (i) State the transition that would give out electromagnetic radiation of longest wavelength.
- (ii) Calculate the longest wavelength of the emitted electromagnetic radiation in (i).

[4]

Q21:JUN 2017 P5

- 2 (b) (i) Describe the photoelectric effect.
- (ii) Calculate the maximum kinetic energy of photoelectrons produced when zinc is irradiated with ultra violet radiation of frequency 4.2×10^{15} Hz, given that the work function of zinc is 4.2 eV.

[4]

- (c) Fig. 2.1 shows some of the energy levels for atomic mercury.

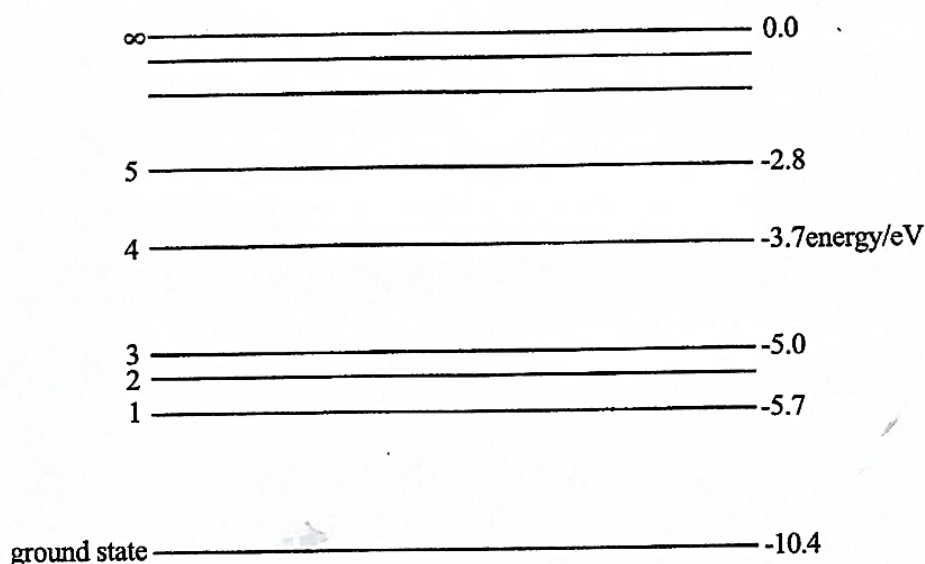


Fig. 2.1

- (i) A transition from energy level 2 to energy level 5 requires absorption of 2.7 eV of energy.

Calculate in electron volts the energy for level 2.

- (ii) An electron falls from energy level 4 to the ground state.

Calculate the frequency of the radiation emitted.

[3]

- (d) Astrophysicists can identify the gases found in the stars by observing and analysing line emission spectra from stars.

Suggest a reason why this identification is effective.

[1]

Q22:NOV 2015 P5

- 1 (d) (i) Explain the term *photon*.
- (ii) Estimate the energy of a red light photon.
- (iii) Deduce the region of the electromagnetic spectrum to which a photon of energy $1 \times 10^{-25} \text{ J}$ belongs.

[5]

Q23:NOV 2012 P5

- 1 (b) (i) Write down Einstein's photoelectric equation.
- (ii) Study the circuit shown in Fig.1.1 and answer questions that follow.

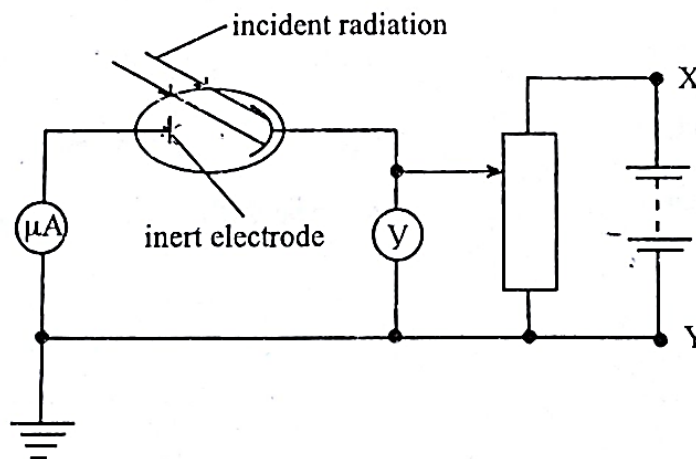


Fig.1.1

- State the effect of increasing the intensity of the incident radiation.
- Suggest how this apparatus could be used to determine the maximum kinetic energy of the photoelectrons.
- The frequency of the incident radiation was varied and the maximum kinetic energy, $K.E_{\text{max}}$, of the photoelectrons was found. Fig. 1.2 shows the results obtained.

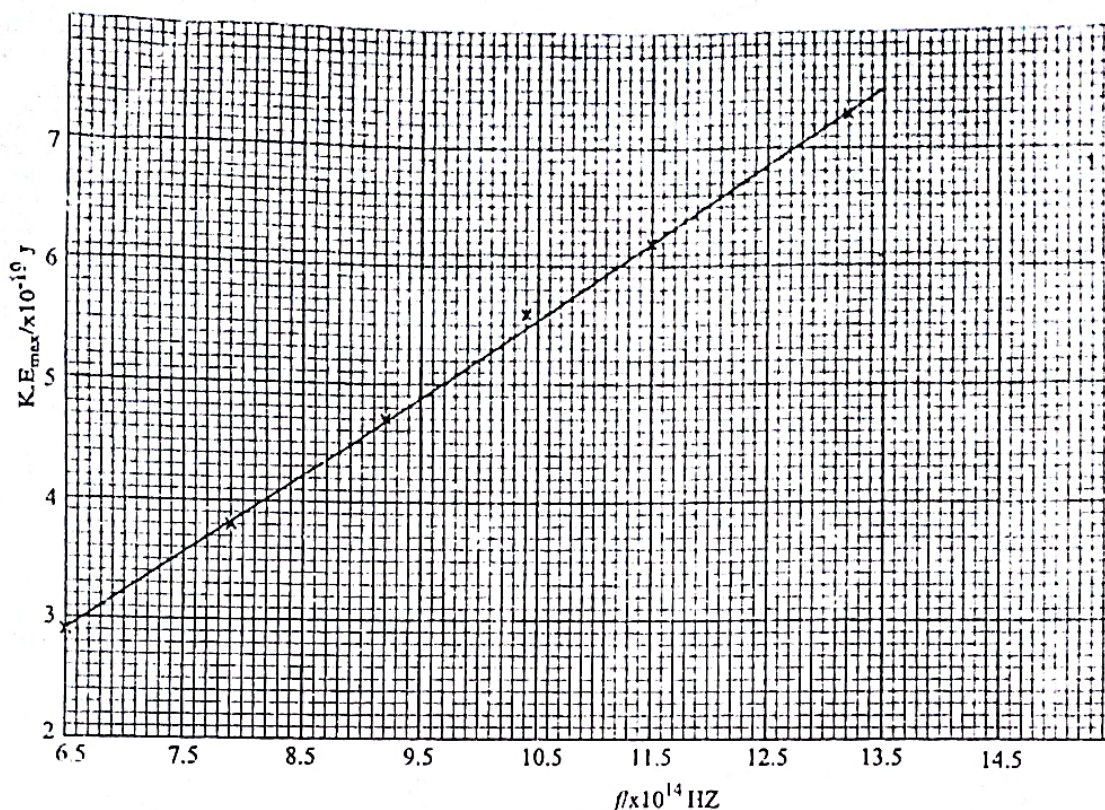


Fig. 1.2

Use Fig 1.2 to determine the Planck constant, the threshold frequency and the work function of the metal.

[13]

Q24:JUN 2012 P5

- 1 (c) (i) Distinguish between *emission* and *absorption* line spectra.
- (ii) Explain the formation of emission line spectra by gases at low pressure.
- (iii) Fig. 1.2 shows some of the energy levels in a hydrogen atom.

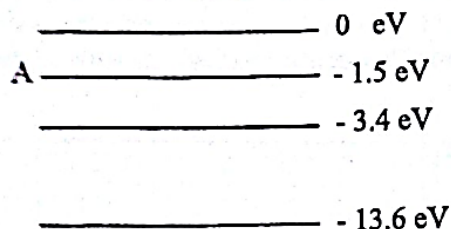


Fig. 1.2

[8]

Calculate the energy in joules which an electron of the hydrogen atom must gain for transition from ground state to energy level A.

Q25:JUN 2018 P5

- 1 (c) (i) Fig. 1.2 shows some energy levels in an atom.

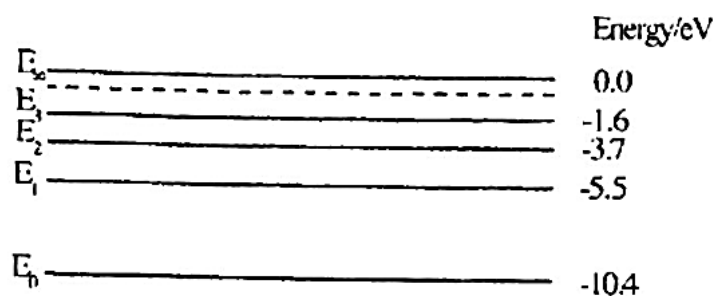


Fig. 1.2

Determine the

1. wavelength, λ , of photons that can cause electron transition from E_0 to E_{∞} ,
 2. momentum, p , of the photons.
- (ii) Identify the part of the electromagnetic spectrum to which the photons belong.

[5]

Q26:NOV 2002 P3

- 5 (a) (i) Write down Einstein's photoelectric equation and explain the meaning of each term.

[2]

- (ii) Sodium has a work function of 2.3 eV.

Calculate

- 1 its threshold frequency,
- 2 the maximum velocity of the photoelectrons produced when the sodium is illuminated by light of wavelength of $5.0 \times 10^{-7}\text{m}$,
- 3 the stopping potential with light of this wavelength.

[8]

Q27:NOV 2006 P3

- 7 (a) (i) Describe with the aid of diagrams the phenomenon of photoelectric emission.

- (ii) State ways in which

1. the maximum kinetic energy,
2. the number of photoelectrons can be increased in photoelectric emission experiments.

7.2 QUANTUM PHYSICS P2, P3&P5 QUESTIONS

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(iii) Explain the significance of the threshold frequency in photoelectric emission.

(iv) In which way does this effect in (iii) contradicts the wave theory? [10]

(b) In experiments to investigate the photoelectric effect, the maximum kinetic energy can be measured by applying a positive potential to stop the photoelectrons.

The results from one such an experiment are shown in Fig. 7.1.

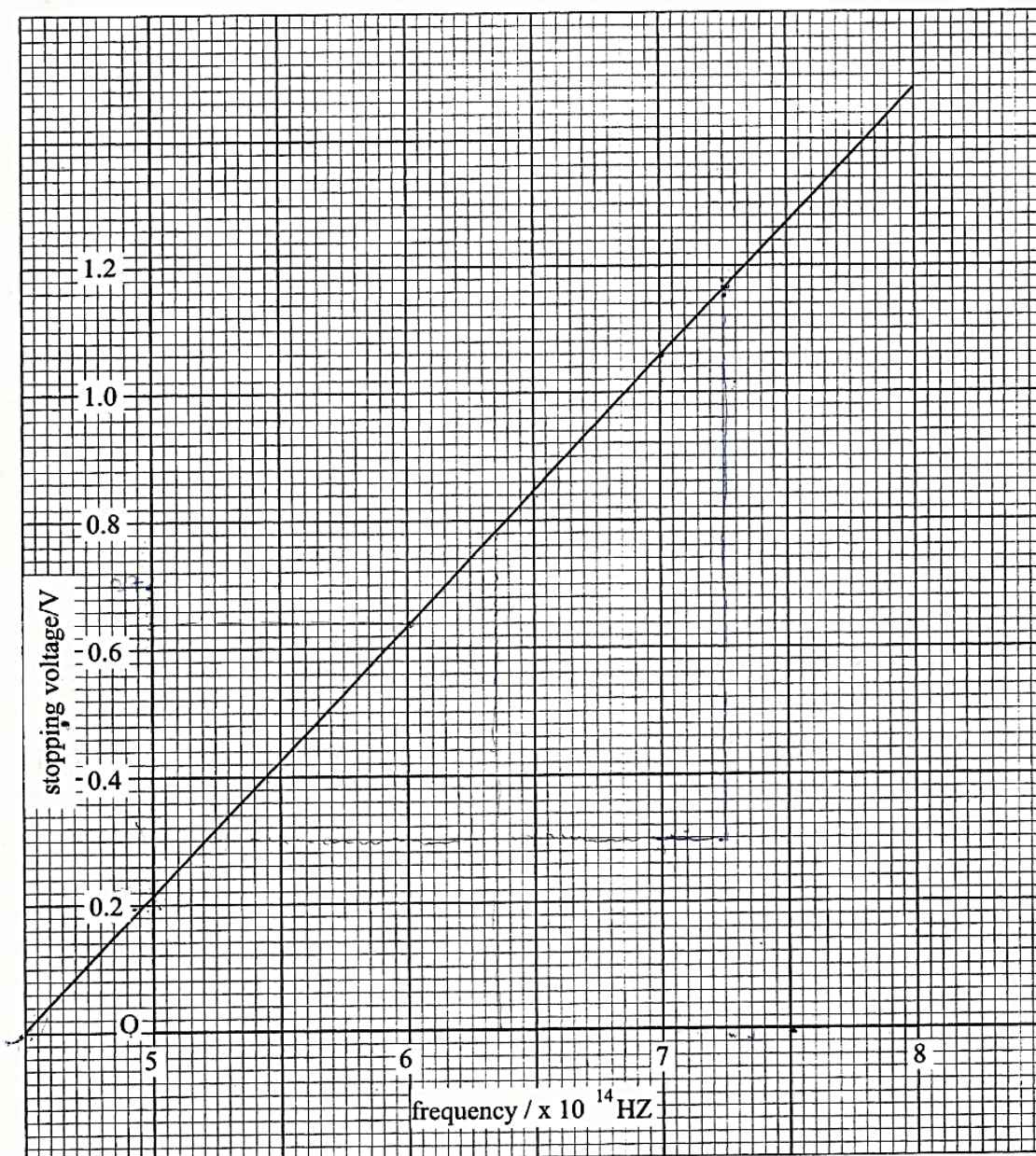


Fig. 7.1

Use Fig. 7.1 to determine

- (i) maximum kinetic energy when the frequency is 6.0×10^{14} Hz,
 - (ii) the Planck's constant, h ,
 - (iii) the threshold frequency and
 - (iv) the work function of the target metal. [6]
- (c) Explain the changes if any that would result to Fig. 7.1 when a target metal of a larger work function is used. [2]
- (d) State one use of a radioactive nuclide with
- (i) a short half-life,
 - (ii) long half-life. [2]

Q28:JUN 2019 P3

- 5 (a) (i) Define the term *photoelectric effect*.
- (ii) The wavelength, λ , of an electromagnetic radiation incident on a metal surface was varied and the corresponding maximum kinetic energy, $E_{\max}$, of the photoelectrons was measured.

Fig. 5.1 shows the corresponding graph of $E_{\max}$ against $\frac{1}{\lambda}$.

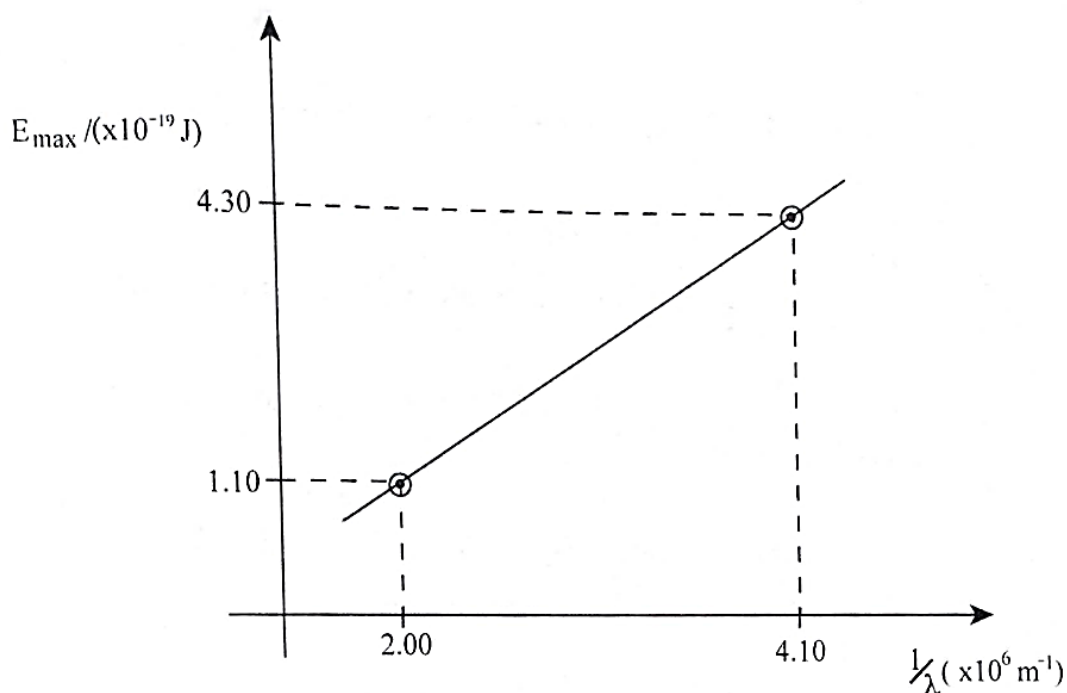


Fig. 5.1

Determine the

1. gradient of the graph,
2. work function of the metal.

[6]

(b) White light is incident on a transparent vessel enclosing cool mercury vapour. The spectrum of the emergent light consists of a number of dark lines.

(i) Account for the dark lines.

(ii) Fig. 5.2 shows some energy levels in an atom of mercury vapour.

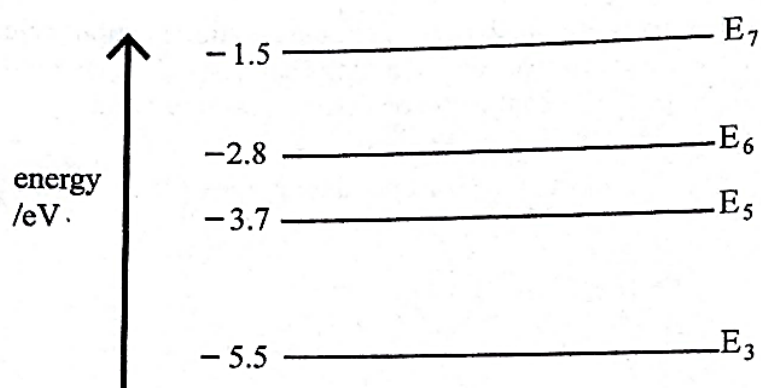


Fig. 5.2

1. Calculate, in eV, the energy of a photon of wavelength 565 nm.
2. Deduce from Fig. 5.2 the transition that gives rise to the energy in b(ii) 1.

[7]